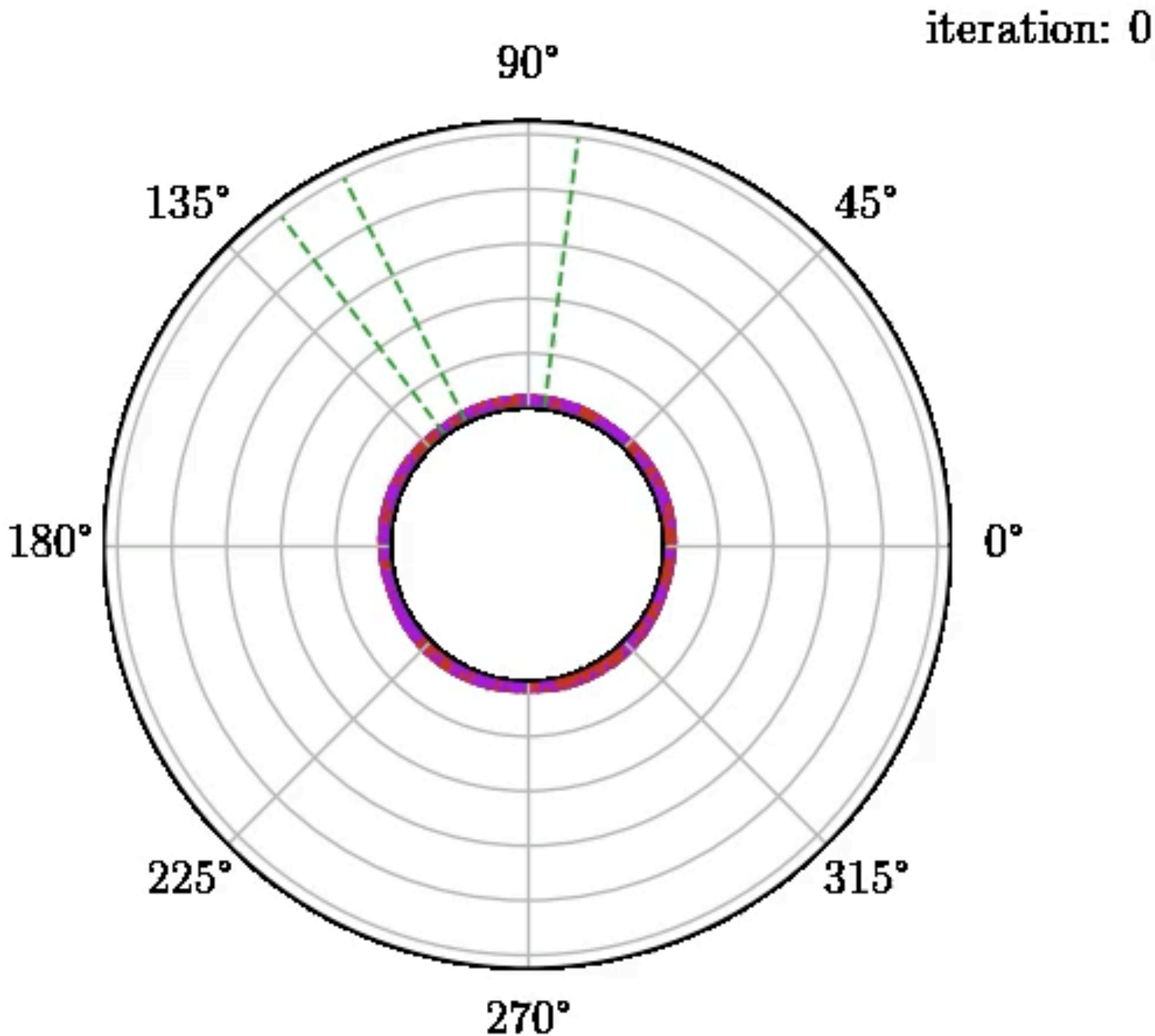
Training dynamics

26

Phase 1: positive neurons align towards critical point D

Phase 2: growth of neurons

Phase 3: do not move from linear regression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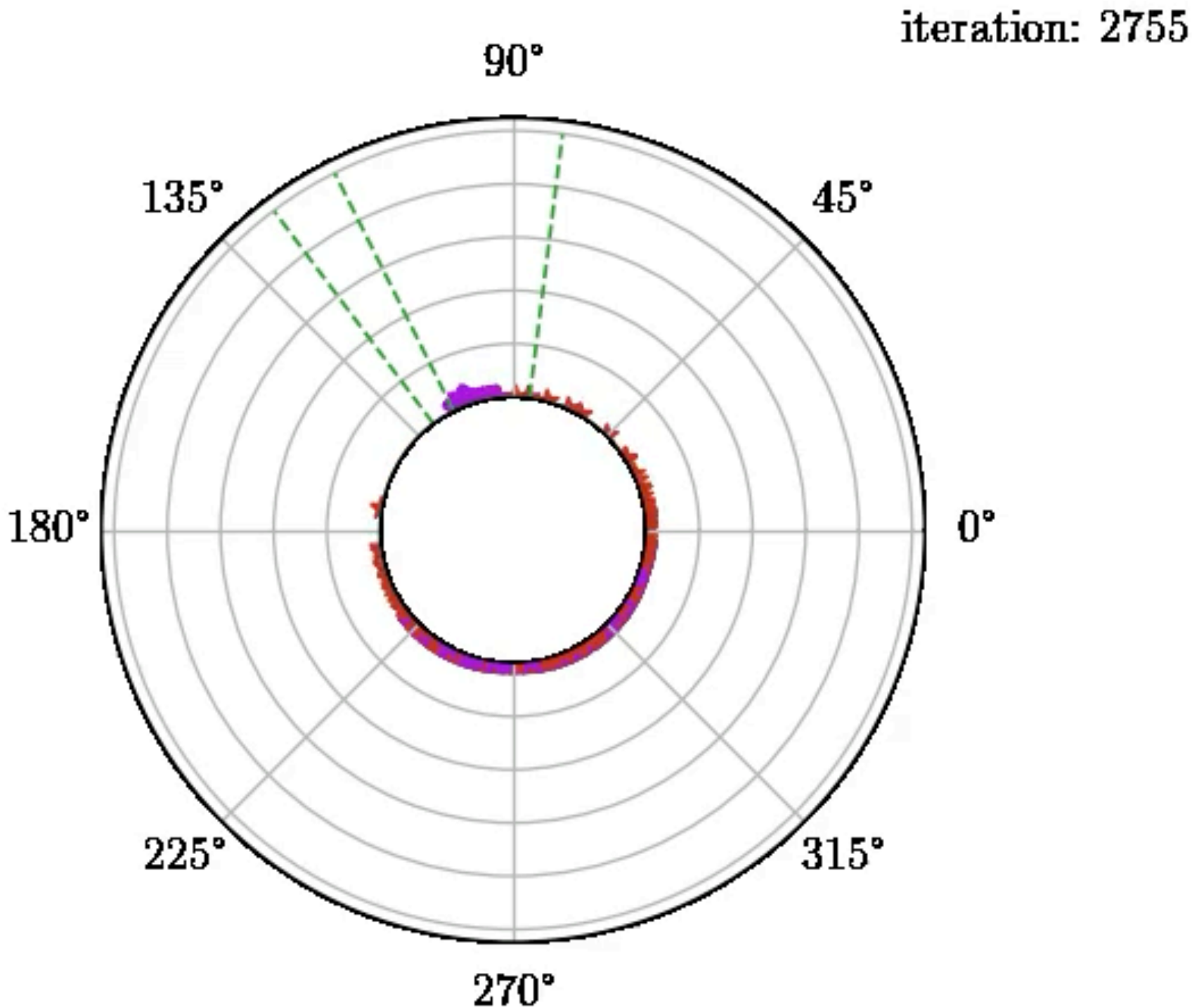
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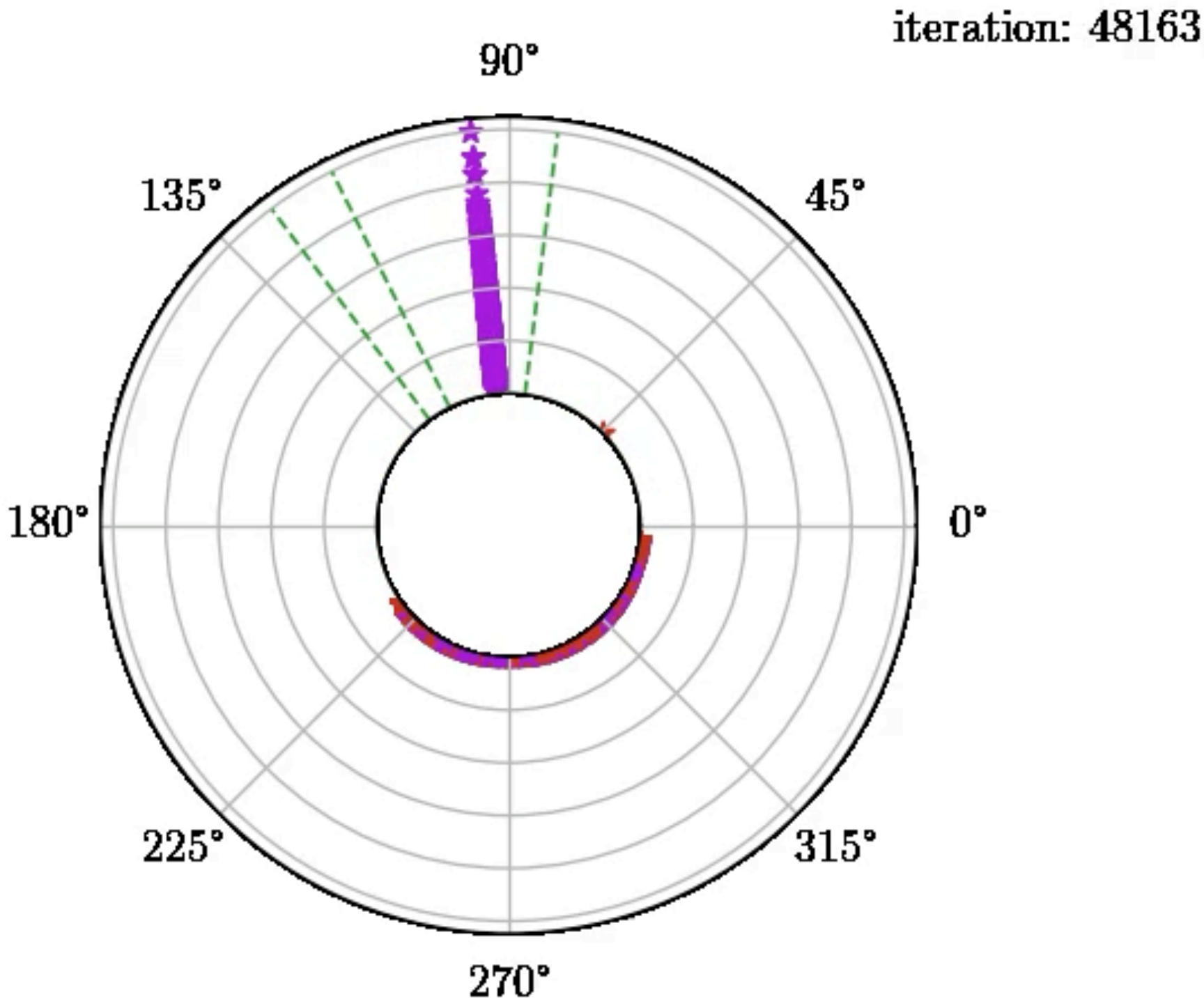
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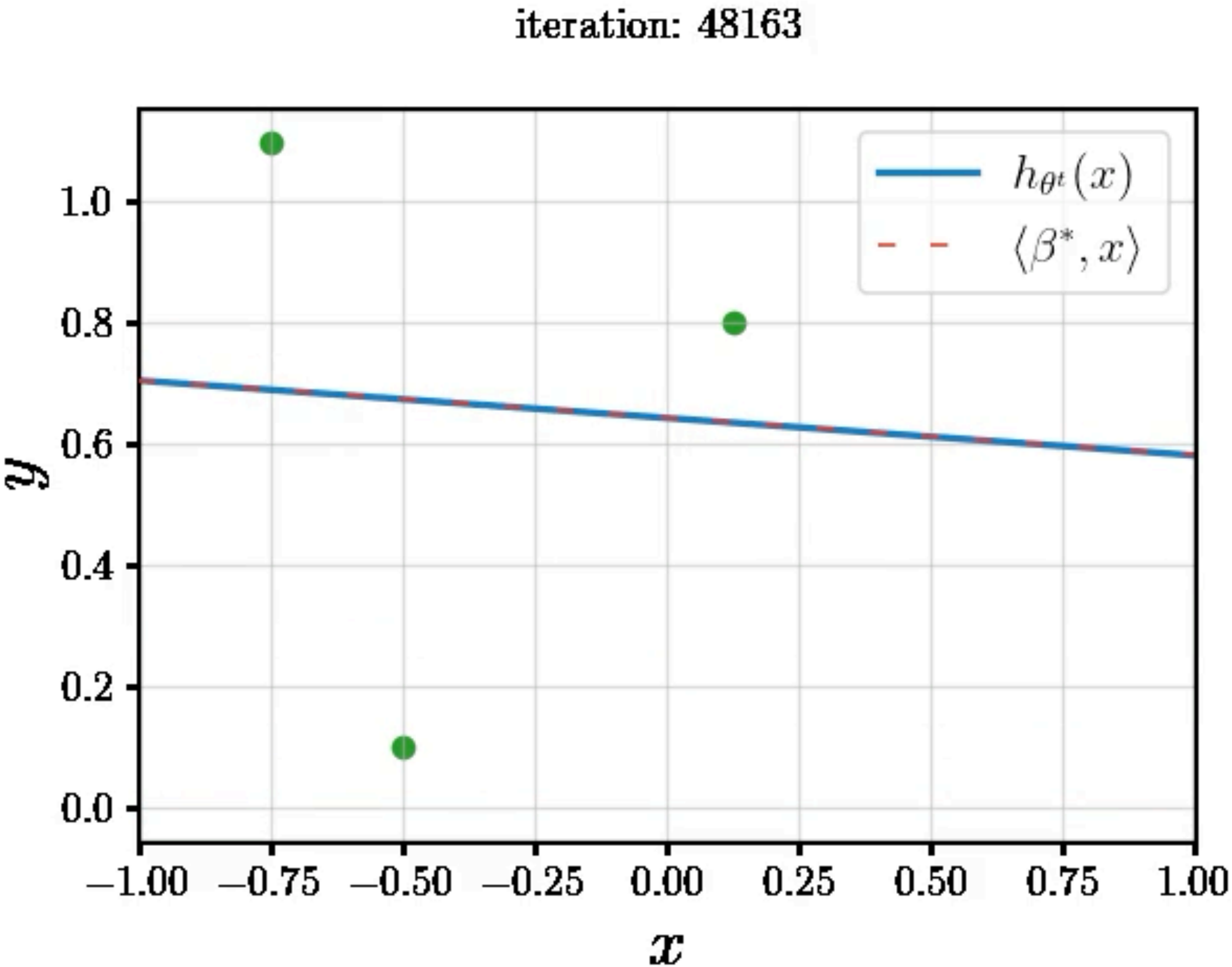
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0.6
0.4
0.2
0.0
y
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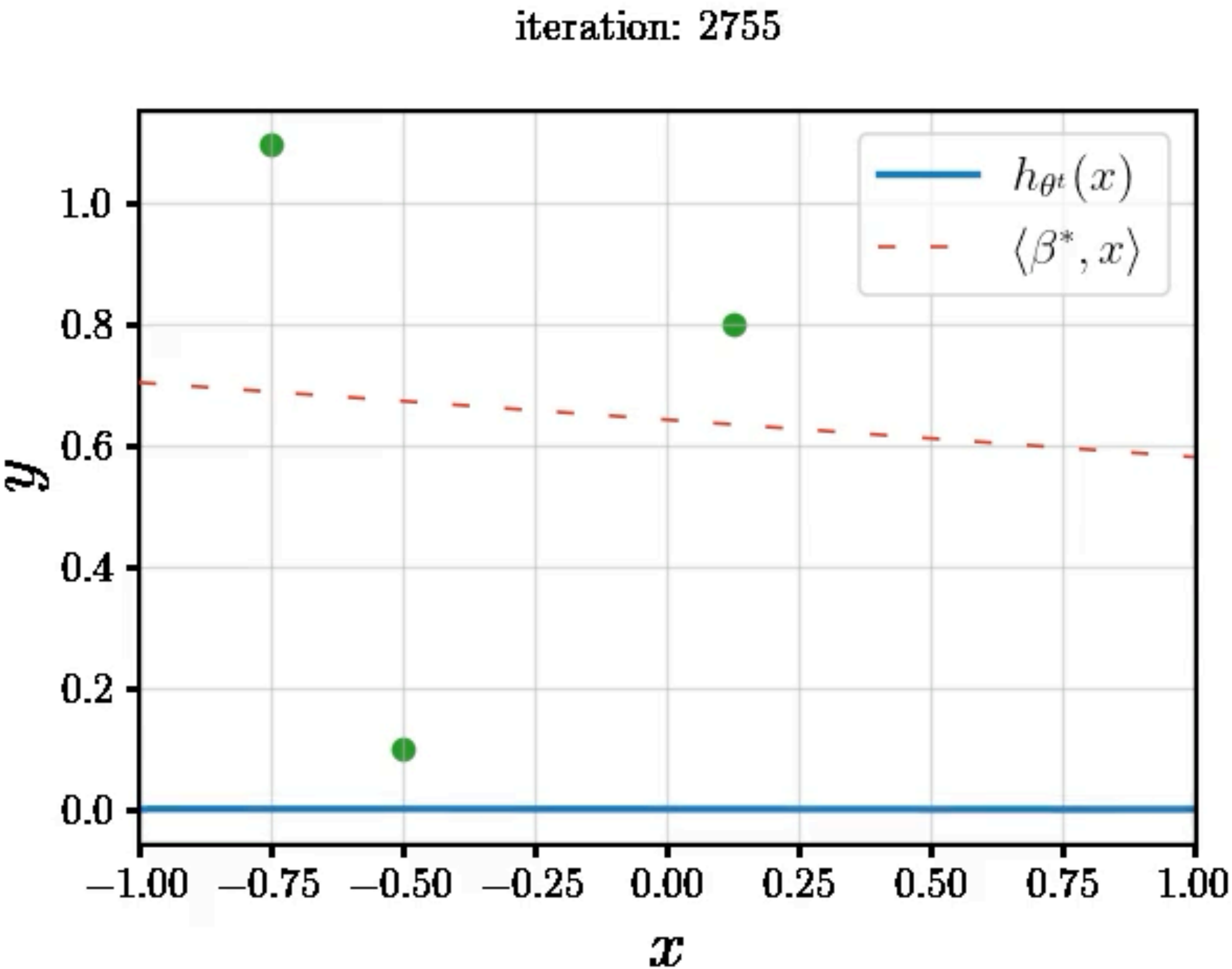
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
x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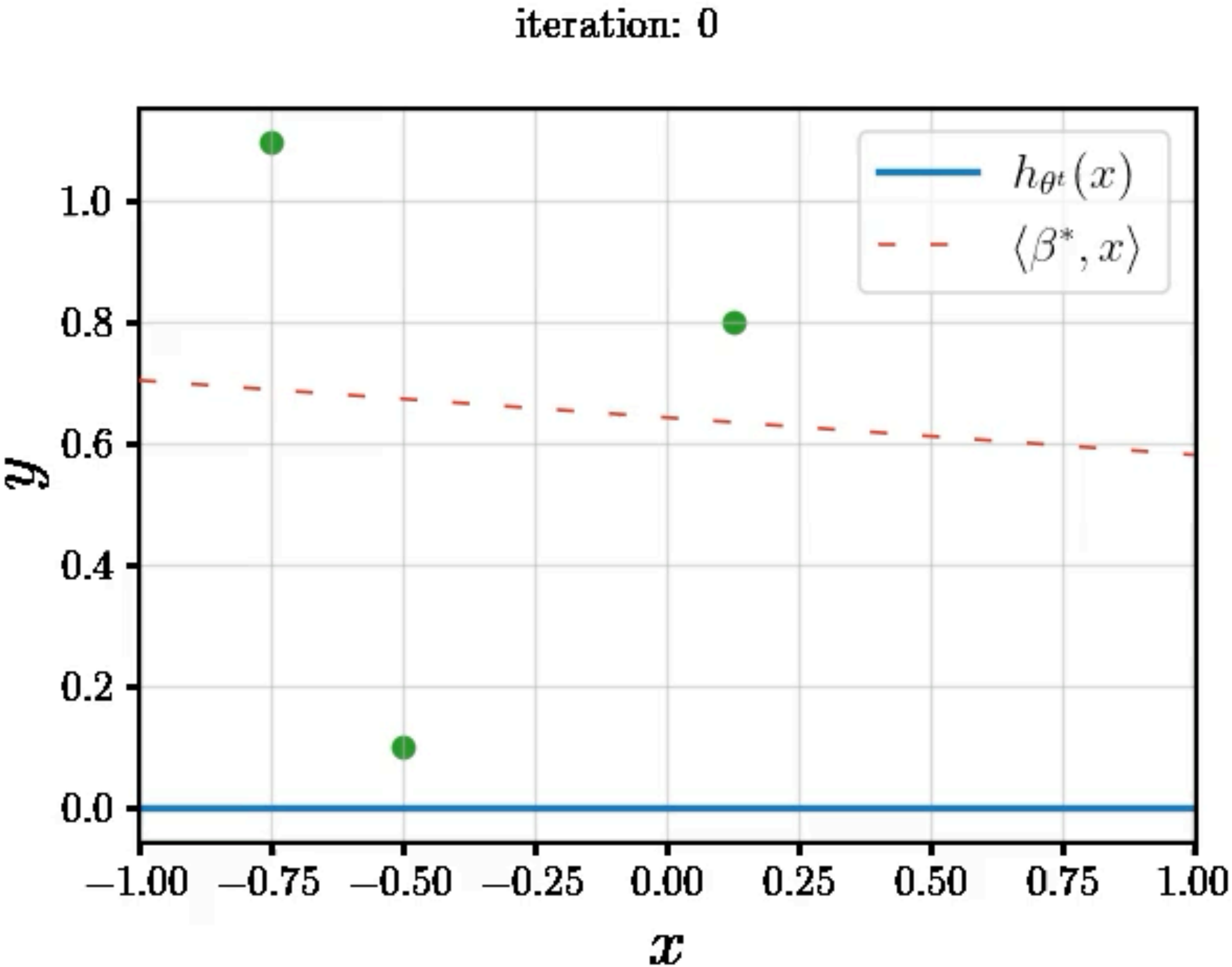
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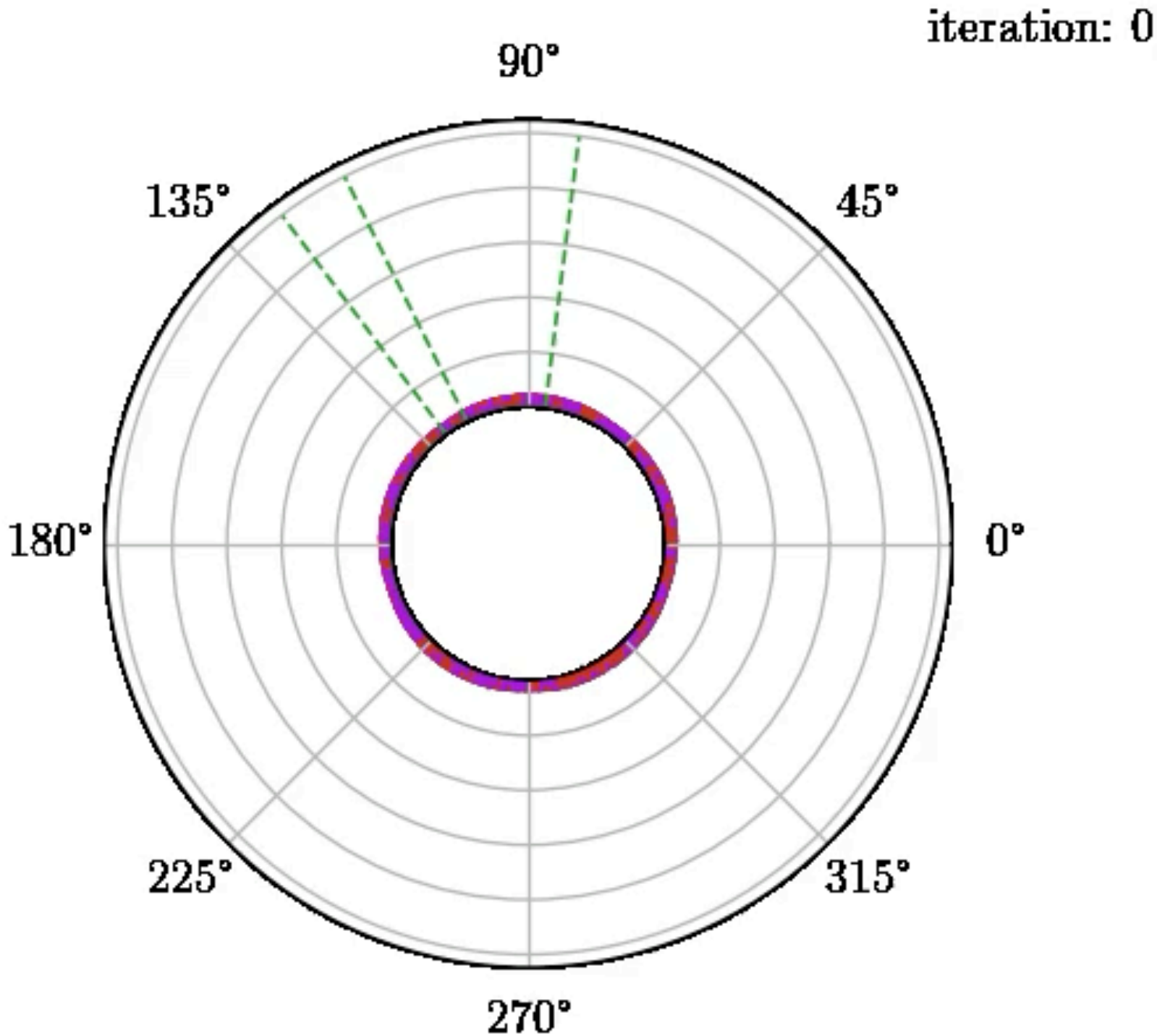
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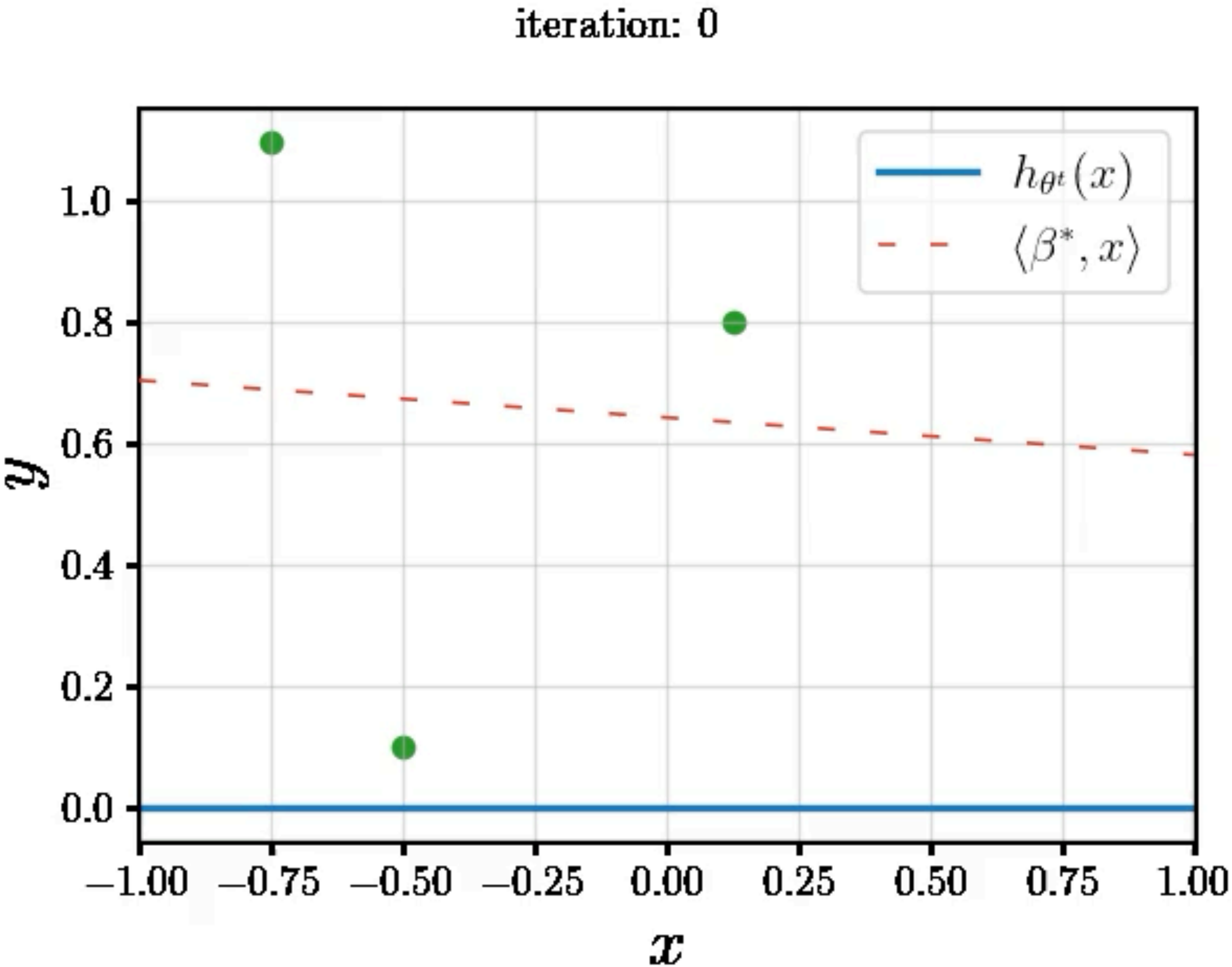
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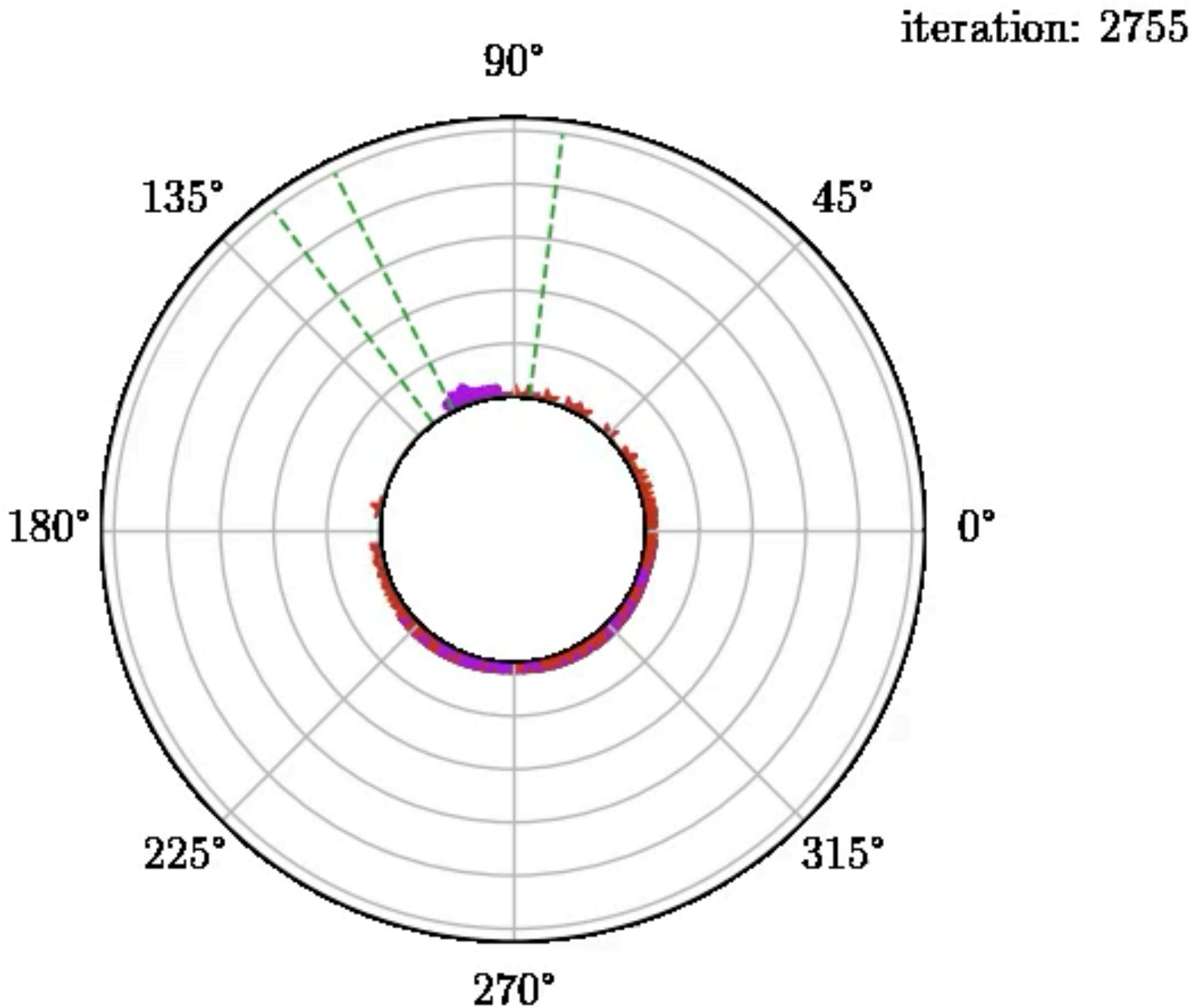
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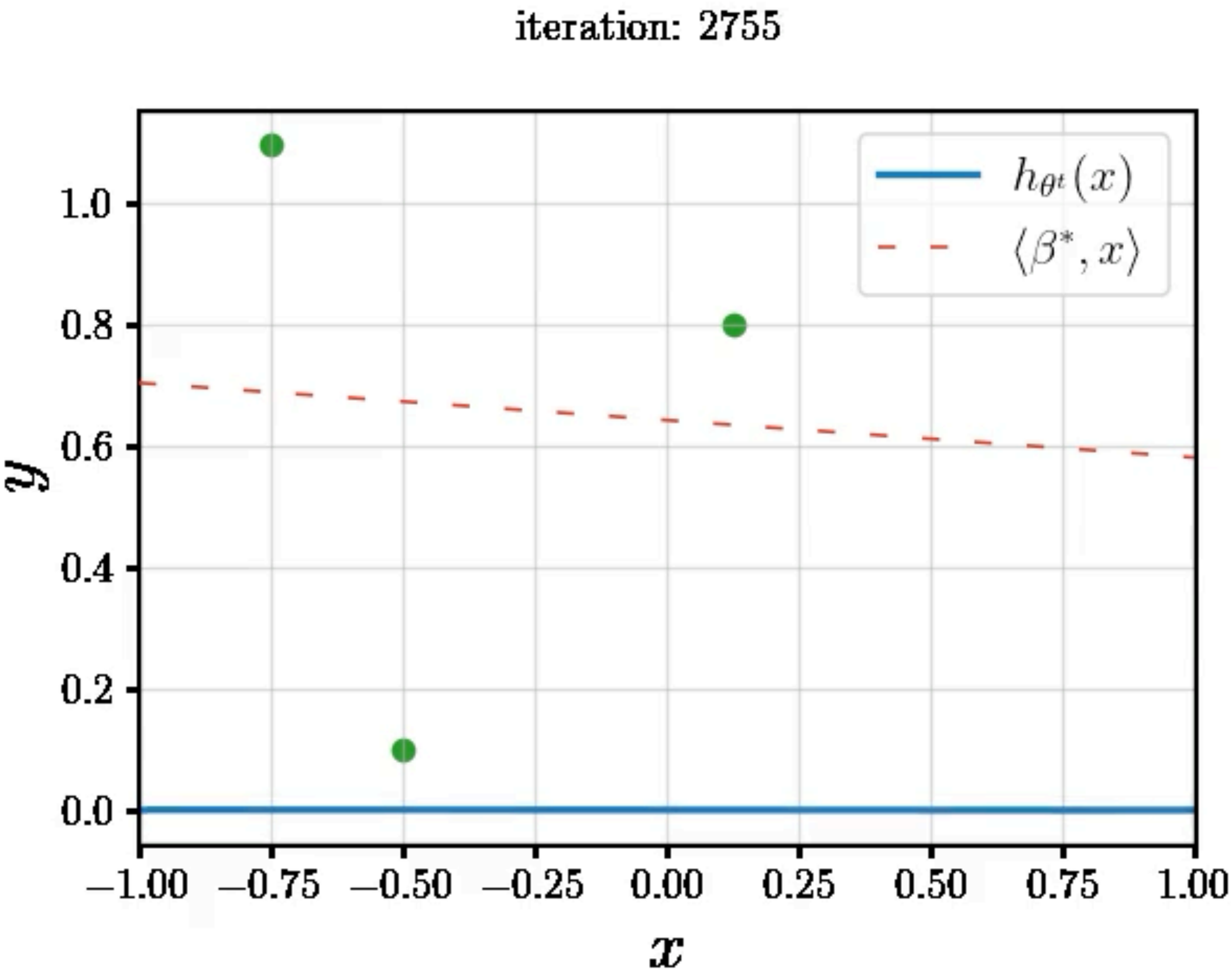
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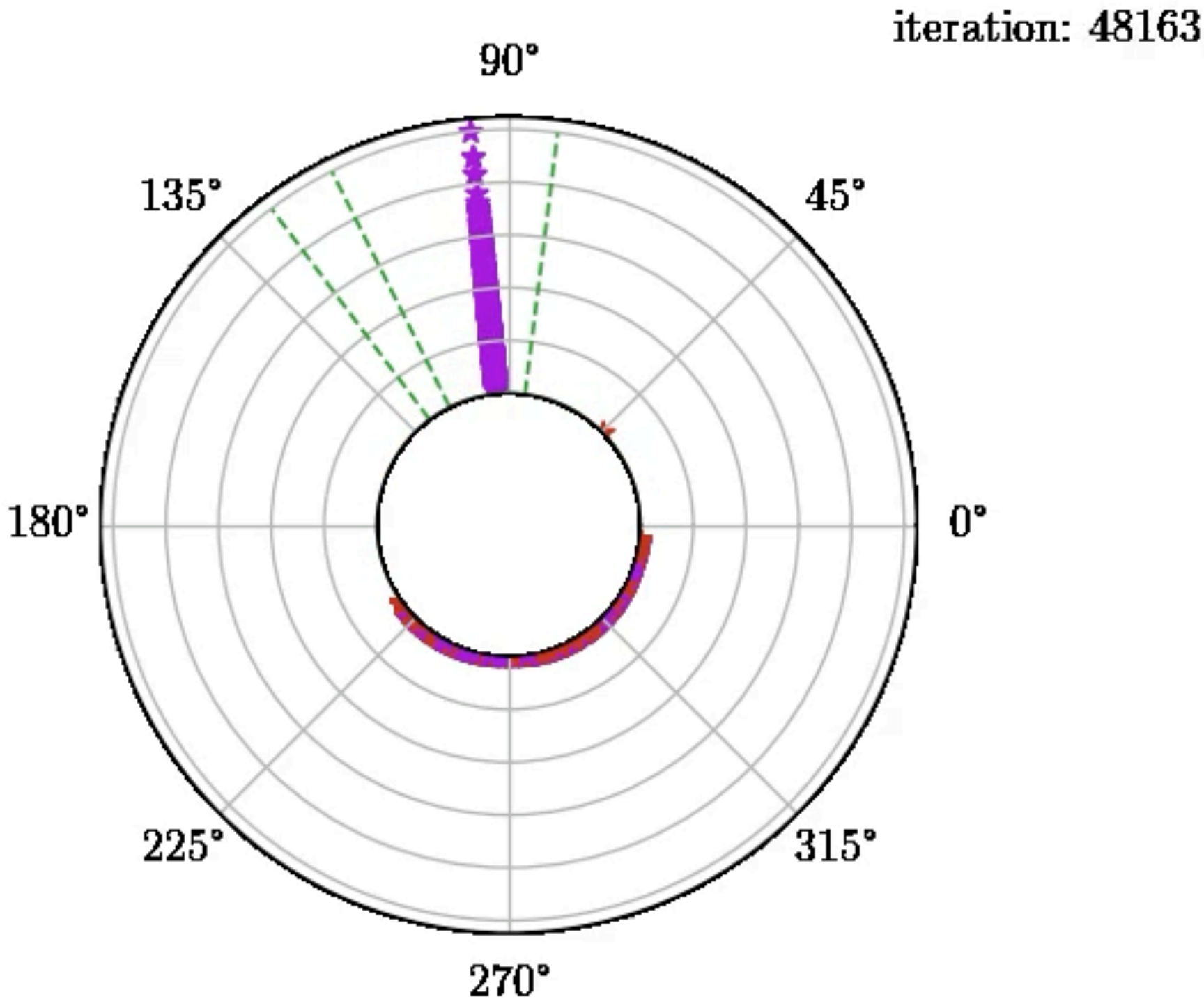
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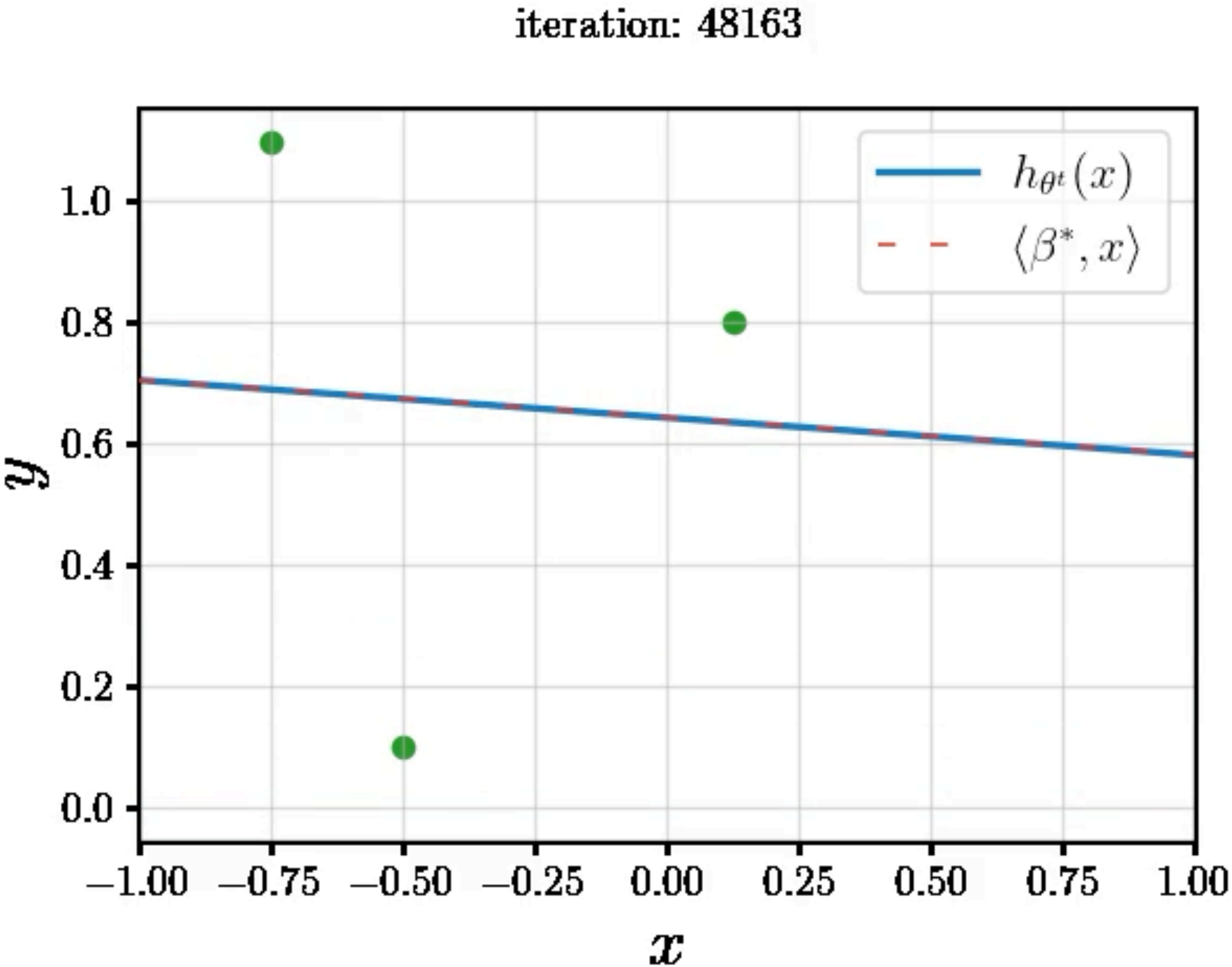
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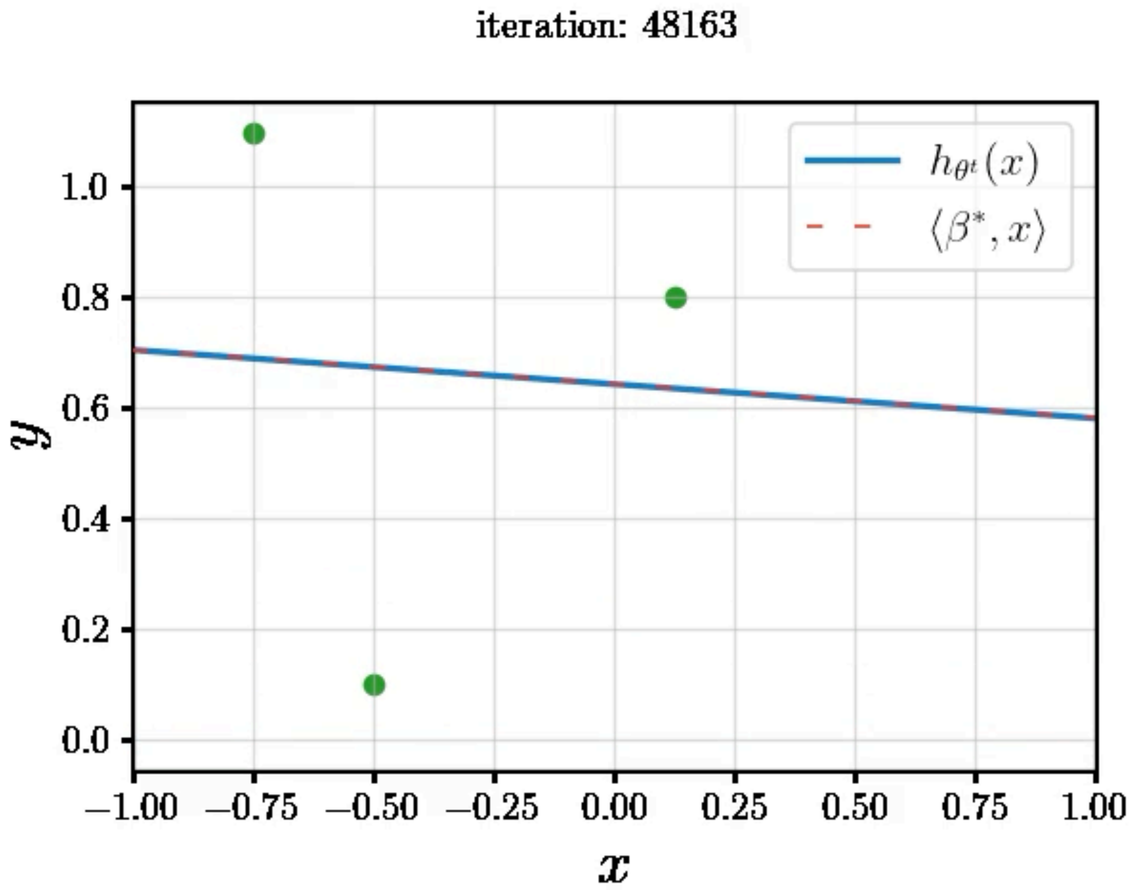
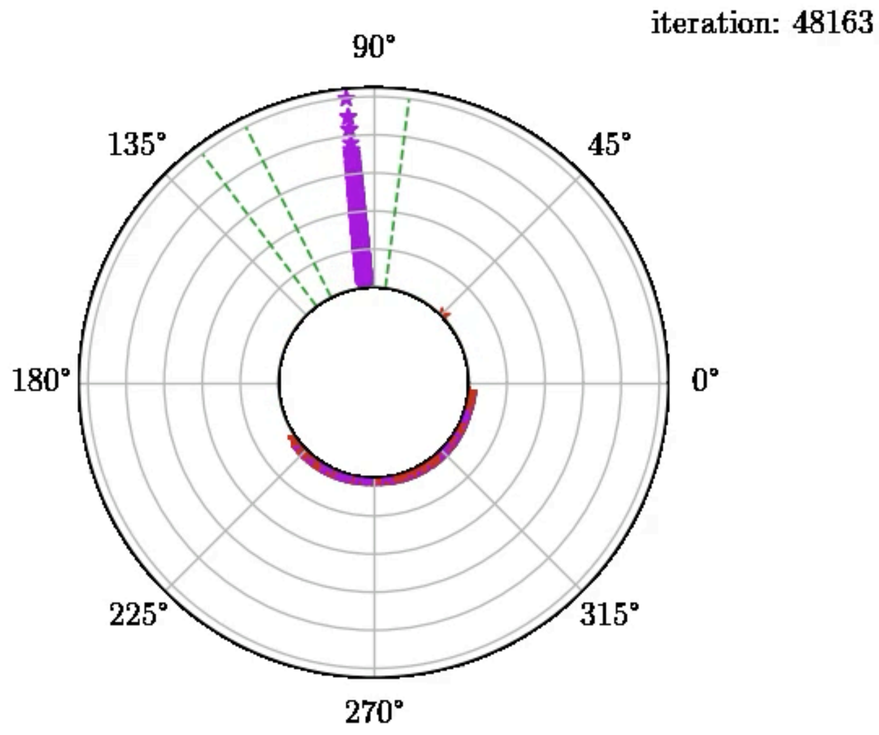
Training dynamics





Phase 1: positive neurons align towards critical point D

Phase 2: growth of neurons

Phase 3: do not move from linear regression

Conclusion

small initialisation leads to early alignment phase

✔ learns sparse features (implicit regularisation)

✘ can lead to spurious local minima (even when $m \to \infty$)

Is ReLU activation a problem?

→ preserving omnidirectionality of neurons seems essential